

Generating Function

The generating function for the sequence $a_0, a_1, a_2, \dots \dots \dots a_r \dots$ Of real numbers is the infinite series

$$\begin{aligned} G(x) &= a_0 + a_1x + a_2x^2 \dots + a_rx^r + \dots \\ &= \sum_{r=0}^{\infty} a_rx^r \end{aligned}$$

Generating Function

Example: The generating functions for the sequences $\{a_r\}$ with

$$a_r = 3 \text{ is } \sum_{r=0}^{\infty} 3x^r$$

$$, a_r = r + 1 \text{ is } \sum_{r=0}^{\infty} (r + 1)x^r ,$$

$$\text{and } a_r = 2^r \text{ is } \sum_{r=0}^{\infty} 2^r x^r$$

Generating Function

The generating function for the finite sequence
$$G(x) = a_0 + a_1x + a_2x^2 + \cdots a_nx^n = \sum_{r=0}^n a_rx^r$$

Example: What is the generating function for the sequence 1,1,1.....,1 is

$$1 + x + x^2 + \cdots + x^r + \cdots = \sum_{r=0}^{\infty} x^r = \frac{1}{1-x}$$

Generating Function

Example: What is the generating function for the sequence $1, -1, 1, -1, \dots, 1$ is

$$1 - x + x^2 - \dots + x^r + \dots = \sum_{r=0}^{\infty} (-1)^r x^r = \frac{1}{1+x}$$

Example: What is the generating function for the sequence $1, a, a^2, a^3, \dots$ is

$$1 + ax + a^2x^2 + \dots + a^r x^r + \dots = \sum_{r=0}^{\infty} a^r x^r = \frac{1}{1-ax}$$

Recurrence Relation

A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence;

Example: $S=\{5,8,11,14,17, \dots\}$ is

$$a_n = a_{n-1} + 3, n \geq 2 \text{ with the initial condition } a_1 = 5$$

Recurrence Relation

Find the first four terms of each of the following recurrence relation.

(i) $a_r = 2a_{r-1} + k$, for all integers $r \geq 2$, $a_1 = 1$

(ii) $a_r = 2a_{r-1} + 3a_{r-2}$, $\forall k \geq 2$, $a_0 = 1$, $a_1 = 2$

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Using Generating functions to solve recurrence Relation

Solve the recurrence relation $a_r = 3a_{r-1}$ for $r = 1, 2, 3, 4 \dots$ and initial condition $a_0 = 2$

$$\begin{aligned}\text{Solution: } a_r &= 3a_{r-1} \\ \Rightarrow a_r x^r - 3a_{r-1} x^r &= 0 \\ \Rightarrow \sum_{r=1}^{\infty} a_r x^r - \sum_{r=1}^{\infty} 3 a_{r-1} x^r &= 0\end{aligned}$$

$$\Rightarrow G(x) - a_0 - 3xG(x) = 0$$

$$\Rightarrow G(x) - 3xG(x) = a_0$$

$$\Rightarrow G(x)(1 - 3x) = a_0$$

$$\Rightarrow G(x) = \frac{a_0}{1-3x} = a_0 \sum_{r=0}^{\infty} 3^r x^r$$

$$\Rightarrow \sum_{r=0}^{\infty} a_r x^r = \frac{a_0}{1-3x} = a_0 \sum_{r=0}^{\infty} 3^r x^r$$

$$a_r = a_0 \cdot 3^r$$

Using Generating functions to solve recurrence Relation

Problem: 1- Solve the recurrence relation

$$a_r - 3a_{r-1} + 2a_{r-2} = 0 \quad \text{for } r = 2, 3, 4 \dots \quad \text{and} \\ \text{initial condition } a_0 = 2, \text{ and } a_1 = 3$$

Problem: 2- Solve the recurrence relation

$$a_r - 2a_{r-1} + a_{r-2} = 2^r \quad \text{for } r = 2, 3, 4 \dots \quad \text{and} \\ \text{boundary condition } a_0 = 2, \text{ and } a_1 = 1$$

Using Generating functions to solve recurrence Relation

Problem: 3- Solve the recurrence relation

$$a_r - 5a_{r-1} + 6a_{r-2} = 2 \quad \text{for } r = 2, 3, 4 \dots \quad \text{and} \\ \text{initial condition } a_0 = 1, \text{ and } a_1 = 1$$

Problem: 4- Solve the recurrence relation

$$a_r - 8a_{r-1} = 10^{r-1} \quad \text{for } a_1 = 9$$